

This command asks for input from the terminal. Now, paste the data of the `id_rsa.pub` file that we copied from the host system. It then appends the data into the `'authorized_keys'` file.

For this to work, SSH has to be installed on the nodes. If not already installed, install SSH and start it using the following commands. Openssh is a software that enables SSH over the systems.

```
$ sudo apt-get install openssh-server
$ sudo service ssh start [6]
```

Now, the Rundeck server can connect to the nodes securely with SSH, and execute the jobs specified for the nodes.

4. Verifying the SSH connection

To check if the Rundeck server is able to log in to the nodes, let us give a test command to be executed on the node. Go to the 'Commands' tab on the Web UI of the Rundeck server, enter the node name (which is 'client' in this case) in the 'Nodes' input text entry, and press 'Search'. Then enter the shell command `'echo test command'` in the 'Command' input text entry. Press `'Run on 1 Node'` to execute the command. The status and the output of the command execution can be seen in Figure 11.

5. Creating a job and running it

In the 'Jobs' tab, click on 'Create a new Job'. We can also do this by clicking on 'Job Actions' and selecting the 'New Job' option from the drop down menu.

Add the job name and the description. In the workflow, we can add as many steps as required, which will be executed sequentially. Here, we will add one step which is to be executed through the command line. Add the command `'echo hello world'`, then save the workflow and also save the job. We can also define files/programs that can be run on the nodes. We can also use the 'Node Filter' option to select the nodes on which we want the job to run. This can be seen in Figures 12 and 13.

Figure 13: The workflow of the job

We can run the job manually using the 'Run Job Now' button, or we can schedule the jobs to run at specified time intervals using 'Run Job Later'.

After the job is run, it gives the status of execution, the time taken to complete the job, and the logs of all the job executions till now. This can be seen in the 'Activity' tab, as shown in Figure 18. In this case, the `'echo hello world'` command is executed on the node and the output `'hello world'` is displayed.

In this way we can deploy the software onto the client nodes

continuously using Rundeck.

The concept of continuous delivery and continuous deployment is attracting a lot of interest from both software developers and researchers. Further topics of discussion could be the integration process of Rundeck with Jenkins (the continuous integration tool) and Logstash (a continuous monitoring tool), scaling up the client nodes using Rundeck, and the security implications of an automated deployment pipeline. **END** 🐧

References

- [1] V. Armenise, "Continuous Delivery with Jenkins: Jenkins Solutions to Implement Continuous Delivery," 2015 IEEE/ACM 3rd International Workshop on Release Engineering, Florence, 2015, pp. 24-27. doi: 10.1109/RELENG.2015.19. URL: <http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=7169448&isnumber=7169432>
- [2] S. A. I. B. S. Arachchi and I. Perera, "Continuous Integration and Continuous Delivery Pipeline Automation for Agile Software Project Management," 2018 Moratuwa Engineering Research Conference (MERCon), Moratuwa, 2018, pp. 156-161. doi: 10.1109/MERCon.2018.8421965. URL: <http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8421965&isnumber=8421884> <https://news.ycombinator.com/item?id=2971521>
- [3] Jenkins, J. 2011. Velocity culture (the unmet challenge in ops). O'Reilly Velocity Conference. URL: <http://assets.en.oreilly.com/1/event/60/VelocityEngine%20Culture%20Presentation.pdf>
- [4] Vivek Gite. 2005. NixCraft. URL: <https://www.cyberciti.biz/howto/question/linux/dpkg-cheat-sheet.php>
- [5] Network Working Group of the IETF, January 2006, RFC 4251, The Secure Shell (SSH) Protocol Architecture
- [6] K.Gaurav. 2018. Rundeck on Ubuntu. URL: <https://blog.koley.in/2018/install-and-setup-rundeck-on-ubuntu>
- [7] M. Shahin, M. Ali Babar and L. Zhu, "Continuous Integration, Delivery and Deployment: A Systematic Review on Approaches, Tools, Challenges and Practices," in IEEE Access, vol. 5, pp. 3909-3943, 2017. doi: 10.1109/ACCESS.2017.2685629. URL: <http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=7884954&isnumber=7859429>

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